

GRIFOLS Diagnostic Grifols, S.A.	Reference:	REGD-0000822	Version:	5.0, CURRENT
	Status:	Effective	Effective Date:	15-Jul-2011
Title:	SM DG57-DG517. Tubing replacement.			Page 2 of 9

DG-517 / DG-517 Service Manual

TUBING REPLACEMENT

Procedure description
This procedure describes some tips when replacing the tubing during Preventive Maintenance procedures. Note: Document REGD-0000822_v5 replaces document REGD-0000822_v4.

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PROCEDURE 1: Syringe racor replacement

EQUIPMENT

1. An instrument switched off.
2. Pliers
3. Cutter
4. Racor
5. Teflon tube (350mm)
6. Silicon tube (20mm)
7. Silicon/Teflon tubes inserting tube
8. A piece of thin sandpaper

SET UP

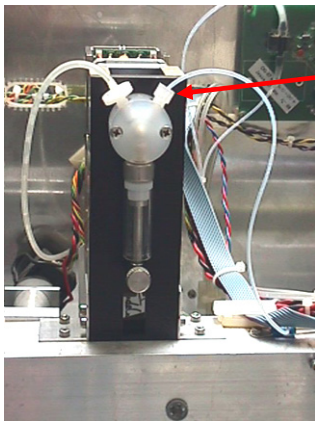
1. ---



WARNING!

There may be rests of blood in the Teflon tube. Perform decontamination procedure REGD-0000787 before changing the tube.

PROCEDURE:

Process	Expected Results
1. Manually unscrew the racor highlighted in Picture 1.  Picture 1 2. We recommend changing the whole Teflon tube going from the syringe to the clot detector. Therefore, remove the tie wraps holding the tube to the clot detector and remove the tube. 3. Cut a piece of Teflon tube a little bit longer than the removed one (final length is 350mm).	--- The tube can be removed from the instrument ---
 WARNING! The use of the cutter may be dangerous.	

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Process	Expected Results
4. Using a cutter, sharpen the edge of the tube as shown in the Picture 2. Picture 2	---
5. Insert the racor to the Teflon tube. 6. Insert the Teflon washer in the Teflon tube as shown in Picture 3. Picture 3	---
7. Completely insert the Teflon tube into the washer by pulling it with pliers while holding the washer with the other hand. See Picture 4. Be aware not to damage the tube. Picture 4	---
8. Using a cutter, cut the Teflon tube leveling it with the washer. 9. Cut a piece of Silicon tube (20mm). See Picture 5.	The racor in the syringe side is ready to be assembled again ---

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Process	Expected Results
Picture 5 10. Insert completely the inserting tool into the Silicon tube. 11. Insert the Teflon tube with the help of a piece of sandpaper. See Picture 6. Be aware not to damage the tube. Insert the tube until it reaches at least half of the Silicon tube. Picture 6 12. Remove the inserting tool 13. Screw 1/8th the syringe racor to the syringe valve (by hand) after it touches-off.. Do not use any tool as the racor may be damaged. 14. Insert the Silicon tube into the clot detector. Place tie wraps to secure the connection. 15. Use document REGD-0000821 to verify this replacement	--- --- --- --- --- ---

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PROCEDURE 2: Probe tubing replacement

EQUIPMENT

1. An instrument switched off.
2. Teflon tube
3. Silicon tube
4. Silicon/Teflon tubes inserting tube
5. A piece of thin sandpaper
6. Allen keys

SET UP

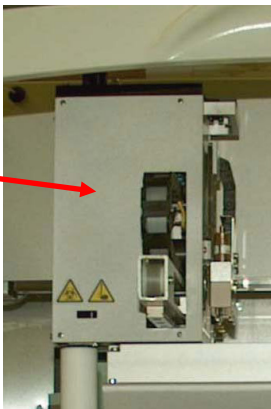
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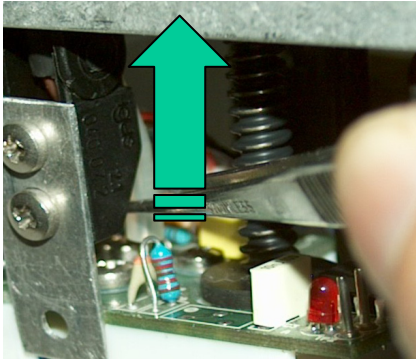
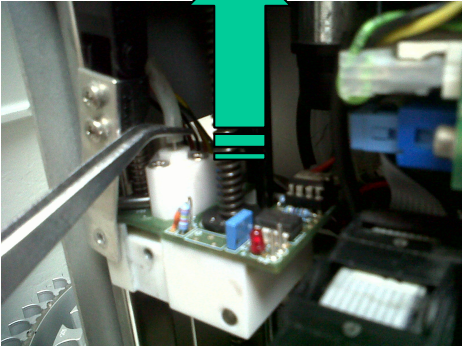
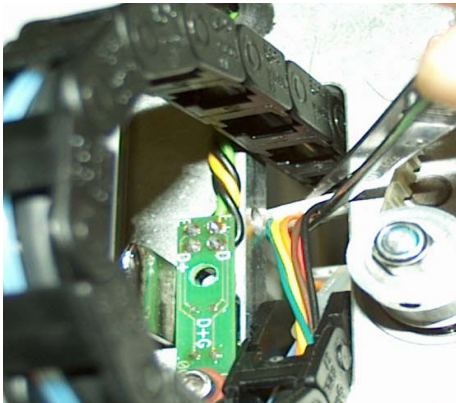
WARNING!

There may be rests of blood in the Teflon tube. Perform decontamination procedure REGD-0000787 before changing the tube.

PROCEDURE:

Process	Expected Results
1. Cut a piece of Teflon of 2000mm.	---
2. Insert a piece (20mm) of Silicon tube in one of the ends. To do it, follow steps 9 to 12 in Procedure 1.	---
3. Remove the tie wraps holding the tube to be substituted to the clot detector.	---
4. Remove the front cover of the Robot Head by unscrewing the 4 Allen screws.	---
 Picture 1	
5. Lower manually the probe. Free the chain by unscrewing the screw highlighted in Picture 2.	---

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Process	Expected Results
 Picture 2 6. Remove the silicon tube using pliers (see Picture 3).  Instrument without impact detector  Instrument with impact detector Picture 3 7. Disconnect the PCB level detector connector (see Picture 4).  Picture 4 8. Remove the silicon tube from the Teflon tube. 9. Attach the end of the tube to be substituted to the free end to the new one using insulating tape. Ensure connection is strong, but thin enough to pass through chains easily.	--- --- --- ---

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DOCUMENT CHANGE HISTORY

Modified connection procedure between probe and Teflon tube and changed document's reference.

Document change history			
Version	Date	Change	Rationale
3.0	2005/05/09	Document REGD-0000822_v3 replaces document 57049V0M001.	Uploaded document to GxPharma.
4.0	2011/04/28	Included in Procedure 2, picture 3 the image of the impact detection hardware. Added references to instrument with and without impact detector hardware.	Picture added due to the new impact detector hardware.
4.0	2011/04/28	Updated the Document Change History: - Date format changed to YYYY/MM/DD - Dates changed to document version effective date.	Adoption of ISO 8601 date format to avoid ambiguity.
4.0	2011/04/28	ID-GELSTATION and PROVUE erased from the title.	Service manual intended for Grifols market.
5.0	See document header.	Document REGD-0000822_v5 replaces document REGD-0000822_v4.	Document version changed.
5.0	See document header.	Step 7 and 13 of procedure 1 modified.	New explanation to ensure proper installation.
5.0	See document header.	Document title.	Commercial reference substituted by project reference.
5.0	See document header.	Document header.	Increase information to easy document search.